

Machine Test 2: Real-Estate Listing Website with AI Integration

Tech Stack

Mandatory

- HTML
- CSS
- JavaScript
- PHP (Core PHP / Laravel / CodeIgniter)
- MySQL

Preferred UI Framework

- Bootstrap 5
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Objective

Build a mini real-estate listing platform with:

- Responsive public listing page
- Property detail page
- AJAX enquiry system
- Secure admin panel
- AI API integration

The platform should allow users to browse properties and send enquiries, while admins can manage listings and enquiries from the backend.

Project Features

1. Home Page

Create a clean and mobile-responsive homepage.

Property Listing Layout

Display properties using:

- Grid layout
- OR
- Card layout

Each Property Card Should Show

- Property Title
 - Featured Image
 - Price
 - Property Type
 - Apartment
 - Villa
 - House
 - Land
 - City
 - View Details Button
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2. Property View Page

Clicking a property should open a dedicated detail page.

Property Detail Page Should Include

- Property Title
- Featured Image
- Price

- Property Type
 - City
 - Full Description
 - Enquiry Button
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3. Enquiry Form

When user clicks the enquiry button:

- Open enquiry form using Bootstrap Modal popup

Fields

- Name
- Mobile Number
- Email
- Message

Requirements

- AJAX form submission
 - Form validation
 - Success/Error messages without page reload
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4. Admin Panel

Create a secure admin panel.

Authentication

- Admin Login
- Logout System

Property Management (CRUD)

Admin should be able to:

- Add Property
- Edit Property
- Delete Property
- View All Properties

Property Fields

- Title
- Description
- Price
- Property Type (Dropdown)
- City
- Featured Image Upload

Enquiry Management

Create a table/page to:

- View all enquiries
 - Property reference
 - Customer details
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5. AI API Integration (Mandatory)

Candidate must integrate at least ONE AI-powered feature using any AI API.

Example AI Features

Choose any one:

Option 1 — AI Property Description Generator

- Admin enters property title/type/city
- AI automatically generates professional property description

Option 2 — AI SEO Meta Generator

Generate:

- SEO title
- Meta description
- Keywords

Option 3 — AI Smart Property Tags

Auto-generate tags like:

- Luxury Villa
- Sea Facing
- Ready To Move
- Investment Property

Option 4 — AI Image Alt Text Generator

Generate image alt text automatically for SEO.

Suggested AI APIs

Developer can use:

- OpenAI API
- Gemini API
- Claude API

Example Usage

- Generate property description
 - SEO content generation
 - Smart tagging
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Technical Requirements

Responsive Design

Frontend must work properly on:

- Desktop
- Tablet
- Mobile

No layout breaking issues.

Bonus Features (Optional)

- Property Type Filter
 - Pagination
 - Featured Property Badge
 - Delete Confirmation Popup
 - Bootstrap 5 UI
 - AJAX-based enquiry form
 - Popup enquiry form using Bootstrap Modal
 - Search by City
 - Image preview before upload
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Database Requirements

Use MySQL database with proper table relationships.

Suggested Tables

- admin_users
 - properties
 - enquiries
 - property_types
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Submission Guidelines

Developer must submit:

- Complete source code
- GitHub repository OR ZIP file
- database.sql file
- README file

README Must Include

- Setup instructions
 - Default admin login credentials
 - AI API setup instructions
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Evaluation Criteria

1. Frontend UI Design
 2. Responsive Design
 3. Backend Functionality
 4. Admin Login System
 5. AJAX Implementation
 6. AI API Integration
 7. Code Structure & Cleanliness
 8. Database Design
 9. Bonus Features
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Important Notes

- Clean code structure is important
- Proper folder architecture should be maintained
- Use reusable components where possible
- Security best practices should be followed
- AI integration should be functional and meaningful

Important

Do NOT use full AI-generated projects. The implementation should reflect the developer's own coding skills and understanding.